



HS 235: Introductory Econometrics

Course Code	HS 235
Course Name	Introductory Econometrics
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG
Distribution	HS Course
Approval	9th Senate (May 2016)

Course Contents

- **Introduction** What is econometrics? [1 Lecture]
- **A Review of some useful statistical tools** Random variables; populations and sample, techniques of sampling and distribution; statistical estimation- estimators and their properties; testing of hypothesis [10 Lectures]
- **Introduction to regression analysis** Assumptions of Classical Linear Regression Models; Ordinary Linear Least Square regression; Gauss Markov theorem; Multiple Regression Analysis- introduction to matrix formulation; interpreting regression coefficients, concepts of residual, fitted value and goodness of fit; Beware of the issues of model misspecifications! Understanding errors in measurement of variables, model selection and non-linear functional forms; Use of Dummy variables [10 Lectures]
- **Violation of assumptions in our model** Multicollinearity, Autocorrelation, Heteroskedasticity: Who are they? How to identify them? What are the causes, effects, and remedies? [12 Lectures]
- **Instrumental variables** Instrumental Variables- their use in solving the problem of omitted variables in regression analysis. [4 Lectures]
- **How econometrics is applied to a real world problem?** Reading and understanding application of econometrics in 1-2 relevant research papers; some basic problem solving- hands-on exercises (preferably in groups). [5 Lectures]

Reading Suggestions

1. Casell, G and Berger, R.L., **Statistical Inference**, 2nd Edition, Duxbury Advanced Series, 2002. (Chapter 2 in particular for statistical distribution)
2. Johnston, J and Dinardo, J., **Econometric Methods**, 4th Edition, McGraw Hill: International Editions, 1997.
3. Wooldridge, Jeffrey M., **Introductory Econometrics**, Thompson, 2003.
4. 1-2 relevant research papers.